

Mini Project : 01

```
#include <stdio.h>

int main() {
    int minutes[7] = {0, 0, 0, 0, 0, 0, 0};
    int i, choice, day, value;
    int total, highest, highestDay;
    float average;
    char confirm;
    FILE *fp;

    // Load previously saved data
    fp = fopen("music_log.txt", "r");

    if (fp != NULL) {
        for (i = 0; i < 7; i++) {
            if (fscanf(fp, "%d", &minutes[i]) != 1) {
                minutes[i] = 0;
            }
        }
        fclose(fp);
    }

    do {
        printf("\n===== MUSIC LISTENING LOGGER ===== \n");
        printf("1. Log listening minutes \n");
        printf("2. View weekly report \n");
        printf("3. Reset weekly data \n");
        printf("4. Exit \n");
        printf("Enter your choice: ");

        if (scanf("%d", &choice) != 1) {
```

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```
printf("Invalid input!\n");
return 1;
}

switch (choice) {

case 1:
    printf("\nEnter day (1=Mon, 2=Tue, ..., 7=Sun): ");
    scanf("%d", &day);

    if (day < 1 || day > 7) {
        printf("Invalid day!\n");
        break;
    }

    printf("Enter listening minutes: ");
    scanf("%d", &value);

    if (value < 0) {
        printf("Minutes cannot be negative!\n");
        break;
    }

    minutes[day - 1] = value;

    // Save all 7 days
    fp = fopen("music_log.txt", "w");

    if (fp == NULL) {
        printf("Error saving file!\n");
        break;
    }
}
```

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```
}
```

```
for (i = 0; i < 7; i++) {  
    fprintf(fp, "%d\n", minutes[i]);  
}
```

```
fclose(fp);
```

```
printf("Listening minutes saved!\n");  
break;
```

case 2:

```
// Read saved data  
fp = fopen("music_log.txt", "r");
```

```
if (fp == NULL) {  
    printf("No saved data found.\n");  
    break;  
}
```

```
for (i = 0; i < 7; i++) {  
    if (fscanf(fp, "%d", &minutes[i]) != 1) {  
        minutes[i] = 0;  
    }  
}
```

```
fclose(fp);
```

```
total = 0;  
highest = minutes[0];  
highestDay = 0;
```

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```
printf("\n===== WEEKLY REPORT =====\n");
```

```
for (i = 0; i < 7; i++) {  
    printf("Day %d: %d minutes\n",  
        i + 1, minutes[i]);
```

```
    total += minutes[i];
```

```
    if (minutes[i] > highest) {  
        highest = minutes[i];  
        highestDay = i;  
    }  
}
```

```
average = total / 7.0;
```

```
printf("\nTotal: %d minutes\n", total);  
printf("Average: %.2f minutes\n", average);  
printf("Highest: %d minutes\n", highest);  
printf("Highest listening day: Day %d\n",  
    highestDay + 1);  
break;
```

case 3:

```
printf("Are you sure you want to reset? (Y/N): ");  
scanf(" %c", &confirm);
```

```
if (confirm == 'Y' || confirm == 'y') {  
    for (i = 0; i < 7; i++) {  
        minutes[i] = 0;
```

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```
    }

    // Clear file contents
    fp = fopen("music_log.txt", "w");

    if (fp == NULL) {
        printf("Error clearing file!\n");
        break;
    }

    fclose(fp);
    printf("Weekly data reset successfully!\n");
} else {
    printf("Reset cancelled.\n");
}
break;

case 4:
    printf("Thank you for using Music Logger!\n");
    break;

default:
    printf("Invalid choice! Try again.\n");
}

} while (choice != 4);

return 0;
}
```

How the program works

1. Program starts

Creates an array of seven days and loads existing values from music_log.txt, if available.

2. Menu appears

The user selects an option from 1 to 4.

3. Log minutes

The user chooses a day and enters listening minutes. The program updates the array and saves all seven values to the file.

4. View weekly report

Reads the file and calculates the total, daily average and highest listening time.

5. Reset or exit

Reset asks for confirmation and clears the saved data. Exit ends the menu loop.

output

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===== MUSIC LISTENING LOGGER =====

1. Log listening minutes
2. View weekly report
3. Reset weekly data
4. Exit

Enter your choice: 1

Enter day (1=Mon, 2=Tue, ..., 7=Sun): 1

Enter listening minutes: 30

Listening minutes saved!

After entering all seven days:

===== WEEKLY REPORT =====

Day 1: 30 minutes

Day 2: 45 minutes

Day 3: 20 minutes

Day 4: 60 minutes

Day 5: 35 minutes

Day 6: 90 minutes

Day 7: 40 minutes

Total: 320 minutes

Average: 45.71 minutes

Highest: 90 minutes

Highest listening day: Day 6